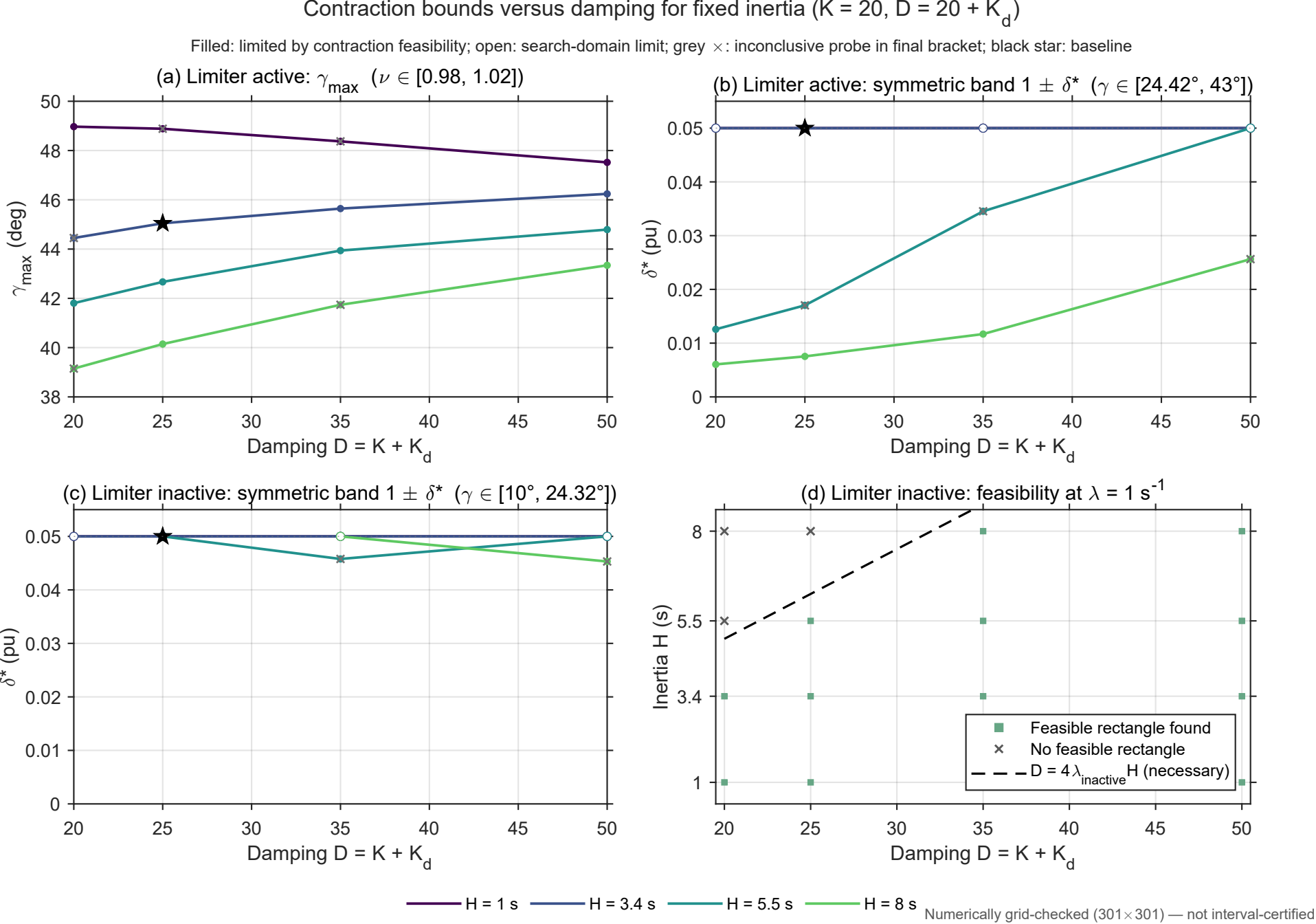
Contraction bounds versus damping for fixed inertia (K = 20, D = 20 + K_d)
Filled: limited by contraction feasibility; open: search-domain limit; grey ×: inconclusive probe in final bracket; black star: baseline
(a) Limiter active: γ_max (ν ∈ [0.98, 1.02])
γ_max (deg)
Damping D = K + K_d
(b) Limiter active: symmetric band 1 ± δ* (γ ∈ [24.42°, 43°])
δ* (pu)
Damping D = K + K_d
(c) Limiter inactive: symmetric band 1 ± δ* (γ ∈ [10°, 24.32°])
δ* (pu)
Damping D = K + K_d
(d) Limiter inactive: feasibility at λ = 1 s^-1
Inertia H (s)
Feasible rectangle found
No feasible rectangle
D = 4 λ_inactive H (necessary)
Damping D = K + K_d
H = 1 s
H = 3.4 s
H = 5.5 s
H = 8 s
Numerically grid-checked (301×301) — not interval-certified